Comparative Impact of ICT on Student Learning Outcomes

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Ed.472: Research Project

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Faculty of Education

BICTE Program

Aadikavi Bhanu Bhakta Campus

Letter of Recommendation

This is to certify that Ms. Sachina Thapa has completed her report entitled Comparative Impact of ICT on Student Learning Outcomes under my supervision as approved by this Department in the prescribed format of the BICTE Program of Department of Education. I hereby recommend her Project Report to be submitted for viva.

…………………………

Bishnu Hadkhale

Report Supervisor

BICTE Program

Date: ________________

Tribhuvan University

Faculty of Education

Aadikavi Bhanubhakta Campus

Letter of Approval

This report, entitled Comparative Impact of ICT on Student Learning Outcomes submitted to the BICTE Program, Department of Education, by Ms. Sachina Thapa has been approved by the undersigned members of the evaluation committee.

Members of Evaluation Committee:

_______________________ Bishnu Hadkhale

Internal Examiner

_______________________ Ghan Bahadur Thapa

HoP

_______________________ Maha Prasad Hadkhale

Campus Chief

Date: _______________________

ACKNOWLEDGEMENT

I would like to express my great sense of gratitude to my respected teacher's irreplaceable guidance who have guided and encouraged me to study Bachelor Degree in BICTE Program and complete this report directly or indirectly. I am very gratitude to Mr. Ghan Bahadur Thapa, the Head of the Program of BICTE for his approval of this report.

My special thanks go to my teacher Mr. Bishnu Hadkhale whose contribution, constructive comments, suggestion and guidance for the completion of the report.

I am equally obliged to my respected Teacher Mr. Maha Prasad Hadkhale, Campus Chief and all the teachers for their proper encouragement. I am directly or indirectly gratitude to all my batch-mates friends, who encouraged and supported me for completion this project report.

Besides, my family members are the key figures who deserve for their moral, ethical and economical support. I thank to my parents who always inspired me for studying also.

………………………..

Sachina Thapa

SELF DECLARATION

I, Ms. Sachina Thapa, hereby declare that the research project entitled "Comparative Impact of ICT on Student Learning Outcomes" has been prepared by me under the supervision of my teacher, Mr. Bishnu Hadkhale, as per the guidelines provided by the Bachelor of Information Communication Technology in Education (BICTE) Program of the Department of Education.

I affirm that this report is my original work, and all the information and data presented have been collected and analyzed by me through surveys and research conducted in three schools of Vyas-2, Damauli, Tanahun. I confirm that proper citations and references have been included for all secondary sources used in the study.

This report is submitted as part of the academic requirement for the BICTE Program and has not been submitted elsewhere for academic credit or recognition.

I take full responsibility for the authenticity and accuracy of the information and findings presented in this report.

Date:

Name: Sachina Thapa

Program: BICTE

Department: Department of Education

Contents

ABSTRACT

This study investigates the transformative role of Information and Communication Technology (ICT) in education, focusing on its impact on academic performance among primary and secondary school students in Vyas-2, Damauli, Tanahun, Nepal. Traditionally, education has been teacher-centered, but the integration of ICT tools such as desktops, laptops, smartphones, and multimedia presentations has shifted the paradigm towards student-centered learning. The research compares the learning outcomes of younger primary students and older secondary students, examining how ICT influences engagement, knowledge retention, and collaborative skills.

Through quantitative data collection from structured questionnaires and academic records, the study reveals that primary students benefit from guided use of ICT, enhancing their engagement and foundational skills, while secondary students exhibit greater independence and critical thinking, utilizing advanced ICT tools to explore complex subjects. The findings highlight significant differences in how these two groups interact with technology, indicating the need for tailored ICT integration strategies that cater to their developmental stages.

In conclusion, the research underscores the potential of ICT to bridge educational gaps and foster equitable learning opportunities. It emphasizes the importance of context-specific strategies, teacher training, infrastructure development, and continuous evaluation to optimize the use of technology in education, ultimately empowering students to achieve their full potential in an increasingly digital world.

Comparative Impact of ICT on Student Learning Outcomes

CHAPTER ONE

1.1. INTRODUCTION

ICT has transformed the educational landscape, which has influenced student how to learn. In traditional classrooms, education was predominantly teacher-centered, relying heavily on textbooks and direct instruction. Students typically assumed a passive role in the learning process, with limited opportunities for interaction and self-directed exploration. The integration of ICT tools, including desktops, laptops, smartphones, slides, and multimedia presentations, has revolutionized this model. These tools have shifted the focus from teacher-driven instruction to a student-centered learning paradigm, where students actively engage with technology to explore, collaborate, and solve problems independently. ICT has introduced dynamic and interactive ways to learn, enabling a deeper understanding of concepts through visual, auditory, and experiential methods.

This study delves into the impact of technology on students' academic performance, with a particular focus on comparing younger primary school students and older secondary school students. By examining these two age groups, the research aims to assess whether the use of ICT enhances students' learning outcomes and to explore whether its effectiveness differs based on the learners' age and developmental stage. A key objective of the study is to identify factors that may contribute to variations in how ICT influences learning. These factors could include the complexity of ICT tools, the diversity and depth of subjects taught at different educational levels, and the developmental shifts in students' cognitive abilities and learning strategies as they grow older. Primary students may require more intuitive and guided use of technology, while secondary students might leverage more advanced tools to support critical thinking and independent learning.

By uncovering these differences, the research seeks to provide actionable insights into optimizing the use of technology in education. It will offer recommendations for tailoring ICT integration to suit the unique needs of learners at different stages of their academic journey. Ultimately, the goal is to ensure that students across all grade levels can derive equitable and meaningful benefits from technological advancements in schools. This research will help us understand better how technology changes what students learn and will guide how we should use technology in schools in the future.

1.2. BACKGROUND OF THE STUDY

The integration of Information Communication and Technology (ICT) has become an important element in transforming the educational landscape. Over the years, ICT tools have been increasing incorporated into teaching and learning process across various educational levels including primary and secondary levels. As technology become more popular and accessible, its potential to impact student learning has increased. However, primary students may rely more on guided instruction whereas secondary student may get benefits from using ICT. This study aims to investigate the comparative impact of ICT on student learning outcomes at primary and secondary school levels of the school that are located in Vyas-2, Damauli, Tanahun. It focuses on how ICT tools influence academic performance of different levels. ICT tools such as desktops, laptop, smartphone, slides, etc has influenced students with enhancing engagement, motivation and supporting collaborative learning environments. This study seeks to explore how ICT impacts learning outcomes in both primary and secondary education comparing factors such as student engagement, knowledge retention and performance. Also this study focuses on assessing the differences in how students at these two levels utilize ICT tools for learning.

1.3. STATEMENT OF THE PROBLEM

The integration of Information and Communication Technology (ICT) into education has become a significant driver of change, transforming traditional teacher-centered classrooms into dynamic, student-centered learning environments. While ICT tools such as desktops, laptops, smartphones, and multimedia presentations have been shown to enhance student engagement, motivation, and collaboration, their effectiveness may vary across different age groups and educational levels.

Primary school students often depend on guided instruction when using ICT, while secondary school students may benefit from more independent and advanced applications of technology. However, there is limited research examining how ICT impacts learning outcomes differently between these two groups, particularly in terms of engagement, knowledge retention, and academic performance.

In the context of schools in Vyas-2, Damauli, Tanahun, there is a need to understand the comparative influence of ICT tools on the learning experiences of primary and secondary students. This research aims to address this gap by investigating the differential effects of ICT on student learning outcomes, exploring how students at these levels utilize technology, and identifying factors that contribute to these differences. The findings will provide valuable insights for tailoring ICT integration strategies to meet the specific needs of learners at different developmental stages and enhance the effectiveness of technology in education.

1.4. OBJECTIVES OF THE STUDY

The main goal of this study is to understand how integrating ICT in education affects the students learning outcomes by comparing primary level student and secondary level students of Vyas-2, Damauli, Tanahun district, Nepal. Two major objectives of this study are:

i. To determine the effects of ICT on academic performance by comparing primary and secondary level students.

ii. To identify the gap between comparing primary and secondary level students.

1.5. HYPOTHESIS OF THE STUDY

In Vyas-2, Damauli, Tanahun district, Nepal, I believe that impact of ICT on student learning outcome differs between primary and secondary level students. Secondary level students may get more benefits by using ICT tools because they can engage more independently with technologies and also secondary students benefits due to their advanced cognitive development, related subject matter, and familiarity with technology. Primary students engaged in the use of ICT tools will prove to be more engaged and motivated in learning activities than those using traditional learning methods, though the impact will be less than among secondary students. The use of ICT in learning will lead to improvement in the collaborative skills of secondary students more than primary students, because they are familiar using technology in group work and communication.

1.6. DELIMITATION OF THE STUDY

This study focuses on the impact of Information and Communication Technology (ICT) on academic performance, with a specific emphasis on comparing primary and secondary-level students in the context of three schools located in Vyas-2, Damauli. Results may not be generalizable to other regions or context. The delimitations of the study are as follows:

1. Geographic Scope

   The study is restricted to three schools in the Vyas-2 area of Damauli, ensuring a localized perspective on the use of ICT in education. Results may not be generalizable to other regions or contexts.

2. Sample size

   - The research targets two specific groups of students: primary school students and secondary school students.
   - The study does not include pre-primary or higher secondary (college-level) students.

3. ICT Tools

   - The study examines commonly used ICT tools such as desktops, laptops, smartphones, slides, and multimedia presentations.
   - Other forms of advanced technology (e.g., virtual reality, artificial intelligence tools) are beyond the scope of this study.

4. Subject Areas

   The research focuses on general academic performance and does not delve deeply into the impact of ICT on specific subjects, such as science or mathematics, unless explicitly observed in the schools studied.

5. Time Frame

   The study is conducted within a limited time frame, which may affect the depth of longitudinal analysis regarding ICT's long-term impact on learning outcomes.

6. Teacher Involvement

   While teachers' perspectives may be considered to provide context, the primary focus remains on students' academic performance and learning outcomes.

7. Learning Context

   The study primarily observes formal classroom learning activities and may not account for informal learning or extracurricular use of ICT outside school settings.

1.7. SELECTION OF THE STUDY AREA

Schools that lies in Vyas-2, Damauli, Tanahun, Nepal were choosen as the study area for this research study. Primary level students and Secondary level students of Satyawati Secondary School (government school), Vyas Divya Jyoti Boarding School (private school), Bal Mandir Secondary School (government school) were the participants of this research study. The study area was selected based on its relevance to the research objectives, which aim to explore the impact of ICT on academic performance by comparing primary and secondary-level students. Schools located in Vyas-2, Damauli, Tanahun, Nepal were chosen due to the availability of diverse educational settings, including both government and private schools, allowing for a comprehensive analysis of ICT integration in varied institutional contexts.

CHAPTER TWO

2. THEORETICAL LITERATURE REVIEW

School-level perceived ICT competence could moderate the relationship between student-level ICT competence and academic outcomes. Academic performance was strongly correlated with student-level perceived ICT competence in schools with a low level of perceived ICT competence; in contrast, this outcome was not observed in schools with a high level of perceived ICT competence. (Yang, Yang, Lu, & Li, 2023)

The majority of students use mobile phones, computers, the internet/modem, social media, digital cameras, or printers outside of school. The findings again indicated that, ICT usage has improved students' academic performance. (Mensah, Quansah, Oteng, & Nii Akai Nettey, 2023)

The teachers and schools play an important part in developing pupils' ICT competences. (Koen Aesaert, 2015)

The relationship between ICT use and the performance of Grade 9 students is examined and further integrated into a multi-level model including school level factors. The results show that characteristics at school level do play a major role in the integration of ICT into teaching and learning and turn out to be relevant across the educational systems. (Koop, 2016)

It is believed that ICT would bring several benefits to the scholars if it is used under the right guidance. The young learner ought to be equipped with quality learning tools. The rapid growth and improvement in Information and Communication Technology (ICT) has caused the flow of technology in education. (Chitnis, 2019)

The main findings are that using of ICT on increasing determination, resolving ability, curiousness, and educational success and totally on learning of secondary-school students has been effective in a high level and this effectiveness differs among males and females with different averages. (kord, 2013)

The content of the learning subjects of Informatics and ICT in Greek schools, which are included as independent subjects in all primary and secondary school grades, aiming at well-educated graduates to be able to use ICT in an effective, creative, and ethically correct way, and to be prepared to participate actively in the digitally enhanced society. (Gousiou & Grammenos, 2023)

ICT brings widened possibilities for the learning processes that are independent from place and space. ICT also allows more flexible (asynchronous) and more personalized learning. It offers new methods of delivering higher education. (lozano)

It should be noted that ICT is important because it helps us to improve aspects related to school, work, leisure activities, and social activities, more quickly. Thus, it is understood that they are indispensable tools in all aspects of daily life. (Salazar, 2018)

CHAPTER THREE

3.1. METHODOLOGY

To investigate the comparative impact of ICT on student learning outcomes at primary and secondary levels in Vyas-2, Damauli, Tanahun district, Nepal, this research employed quantitative data collection techniques. Quantitative data was gathered through structured questionnaires administered to 20 students, equally divided between primary and secondary levels. These surveys collected information in students' perceptions of ICT's impact on their learning, and their overall academic performance. Additionally, academic records from local schools were analyzed to assess trends in academic performance among students who use ICT tools. The collected information was organized into appropriate tables and charts to align with the research objectives.

3.2. ETHICAL ISSUES

Students participating in the study were briefed on the research objectives, procedures, and potential benefits of the study in language appropriate for their age and comprehension level. Their voluntary agreement (assent) was obtained before data collection, ensuring that they fully understood their involvement. Participants' personal information, such as names and school identifiers, was anonymized during data analysis and reporting. This involves a sensitive approach to minimize distress and ensure informed consent. Maintaining a balanced and unbiased research process is essential, avoiding coercion, personal belief imposition, and ensuring accurate and transparent reporting. Protecting participants' confidentiality through anonymizing data, secure storage, and clear communication about data usage is crucial. Adherence to ethical guidelines, including obtaining ethics committee approval and complying with local laws, is mandatory. Ethical dissemination of findings requires clear and accessible communication, safeguarding anonymity, and using findings responsibly to improve related policies and practices. Upholding these ethical standards ensures the integrity and credibility of the study while respecting the rights and dignity of participants.

CHAPTER FOUR

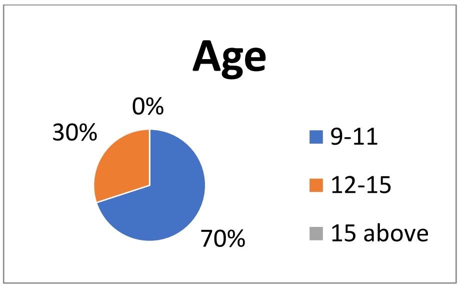
4.1. RESULT AND ANALYSIS OF SURVEY

In this section an analytical interviews had been conducted with the people who were involved in use of ICT in their education.

1.1.1 Description of Results

1. What is your age?

Figure 1



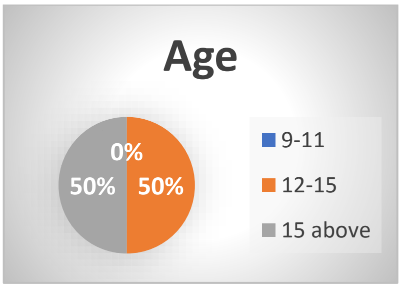
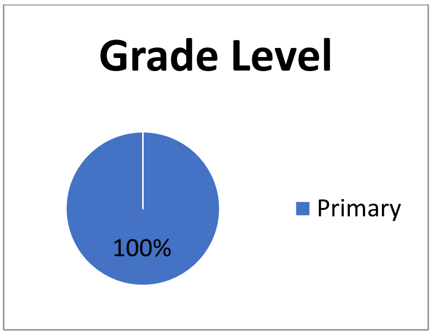


Table 1

9–11 years: 70% of respondents fall into this age group, making it the dominant demographic in the study.	9–11 years: 0% of respondents are in this age range, indicating no participation from young individuals.
12–15 years: 30% of respondents are in this age range, representing a smaller but significant portion.	12–15 years: 50% of respondents are in this age range.
15 above: 0% of respondents are aged 15 and above, indicating no participation from older individuals.	15 above: 50% of respondents are in this age range

2. What is your grade level?

Figure 2



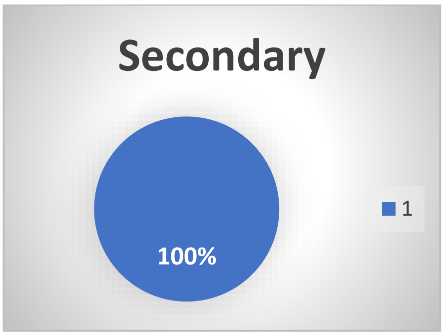
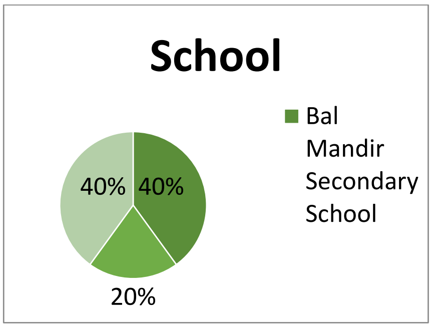


Table 2

Primary: 100% of respondents are at the primary grade level, indicating participation from younger students.	Secondary: 100% of respondents are at the secondary grade level, indicating participation from younger students.

3. What is your school name?

Figure 3



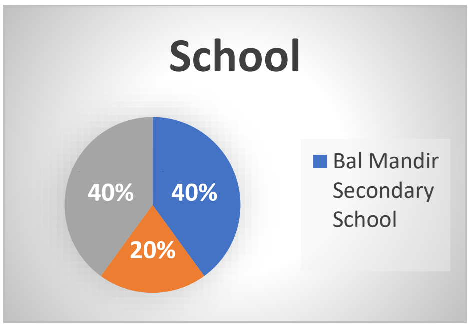


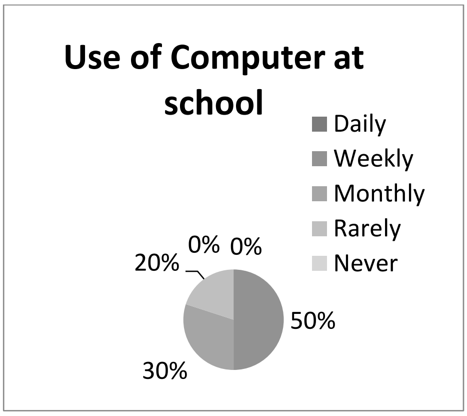
Table 3

Bal Mandir Secondary School: 40% of respondents are from this school, indicating significant representation.	Bal Mandir Secondary School: 40% of respondents are from this school, indicating significant representation.

Vyas Divyajyoti Boarding School: 20% of respondents belong to this school, showing lower participation compared to others.	Vyas Divyajyoti Boarding School: 20% of respondents belong to this school, showing lower participation compared to others.
Satyawati Secondary School: 40% of respondents are from this school, reflecting equal representation with Bal Mandir Secondary School.	Satyawati Secondary School: 40% of respondents are from this school, reflecting equal representation with Bal Mandir Secondary School.

4. How often do you use computer at school?

Figure 4



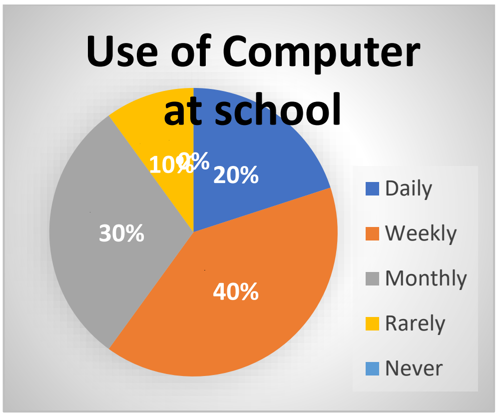


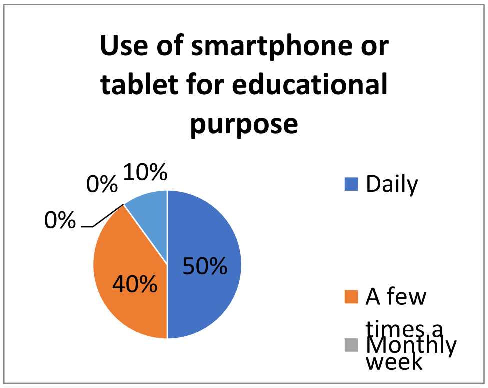
Table 4

Daily Use: 50% of students use computers daily, indicating strong integration into daily school activities.	Daily Use: 20% of students use computers daily showing moderate engagement.

Weekly Use: 30% access computers weekly, showing moderate engagement.	Weekly Use: 40% access computers weekly, indicating strong integration into daily school activities.
Monthly Use: 20% use computers monthly, reflecting periodic interaction.	Monthly Use: 30% use computers monthly, reflecting periodic interaction.
Rarely: 0% reported rare usage, suggesting consistent access among respondents.	Rarely: 10% reported rare usage, suggesting consistent access among respondents.
Never: 0% indicated no usage, emphasizing universal exposure to computers at school.	Never: 0% indicated no usage, emphasizing universal exposure to computers at school.

5. How often do you use a smartphone or tablet for educational purposes?

Figure 5



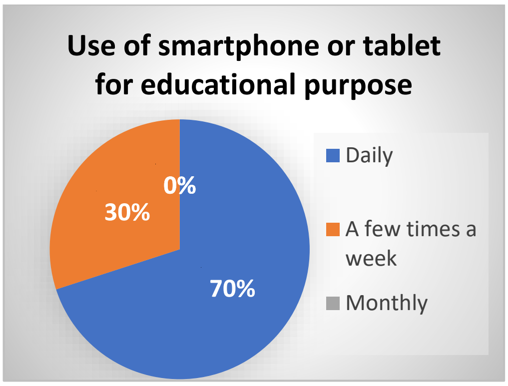


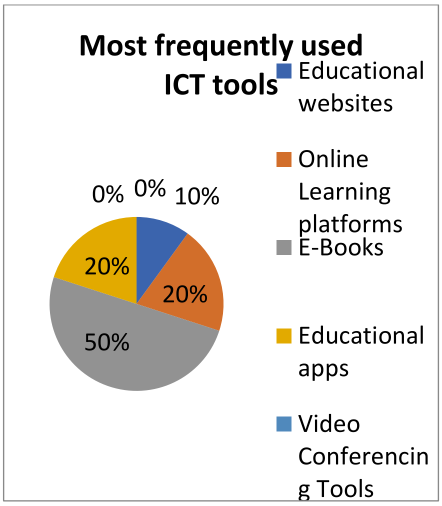
Table 5

Daily Use: 50% of respondents use smartphones or tablets daily, indicating a	Daily Use: 70% of respondents use smartphones or tablets daily,

strong reliance on these devices for learning.	indicating a strong reliance on these devices for learning.
A Few Times a Week: 40% use these devices a few times a week, showing moderate engagement.	A Few Times a Week: 30% use these devices a few times a week, showing moderate engagement.
Monthly Use: 10% access smartphones or tablets for education monthly, reflecting minimal frequency.	Monthly Use: 0% access smartphones or tablets for education monthly, reflecting minimal frequency.
Rarely: 0% reported rare or no usage, suggesting that nearly all respondents integrate these devices into their educational activities to some extent.	Rarely: 0% reported rare or no usage, suggesting that nearly all respondents integrate these devices into their educational activities to some extent.

6. What ICT tools do you use most frequently for learning?

Figure 6



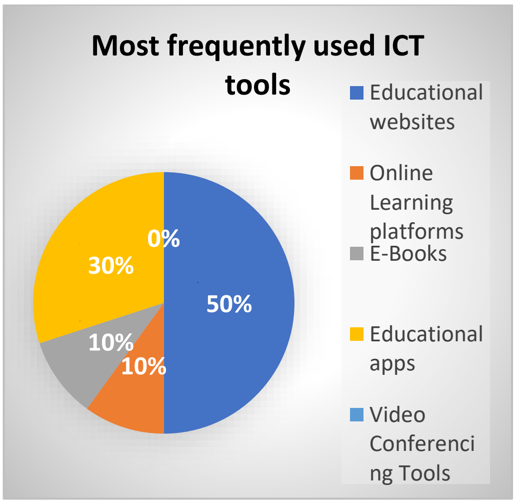
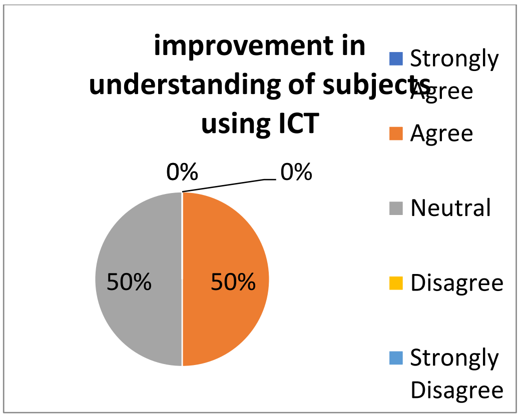


Table 6

Educational Websites: 10% of respondents prefer using educational websites, reflecting limited usage.	Educational Websites: 50% of respondents prefer using educational websites, making them the most popular ICT tool.
Online Learning Platforms: 20% frequently use these platforms, indicating moderate adoption.	Online Learning Platforms: 10% frequently use these platforms, reflecting limited usage.
E-Books: 50% of respondents rely on e-books, making them the most popular ICT tool.	E-Books: 10% of respondents rely on e-books, making reflecting limited usage.
Educational Apps: 20% of respondents use educational apps, showing notable engagement.	Educational Apps: 30% of respondents use educational apps, showing notable engagement.
Video Conferencing Tools: 0% reported using these tools, suggesting minimal relevance in the study context.	Video Conferencing Tools: 0% reported using these tools, suggesting minimal relevance in the study context.

7. Do you think ICT tools have improved your understanding of subjects?

   Figure 7



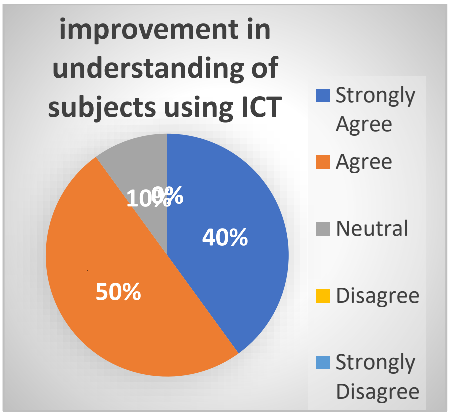
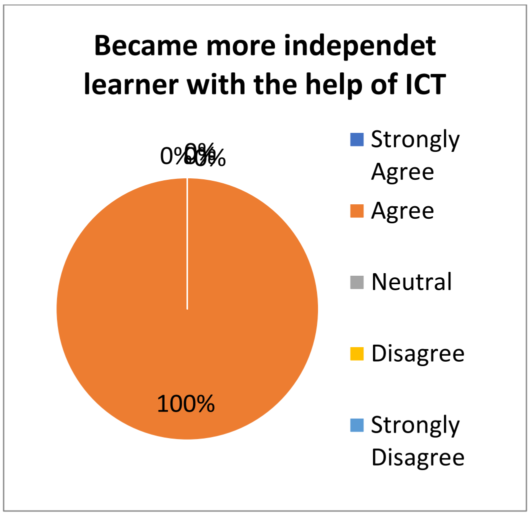


Table 7

Agree: 50% of respondents agree that ICT improves their understanding of subjects, making it the most common response.	Agree: 50% of respondents agree that ICT improves their understanding of subjects, making it the most common response.
Neutral: 50% of respondents are neutral, showing no strong opinion on the matter.	Neutral: 10% of respondents are neutral, showing no strong opinion on the matter.
Strongly Agree: 0% of respondents selected these options, indicating no extreme views.	Strongly Agree: 40% of respondents selected these options.
Disagree: 0% of respondents selected these options, indicating no extreme views.	Disagree: 0% of respondents selected these options, indicating no extreme views.
Strongly Disagree: 0% of respondents selected these options, indicating no extreme views.	Strongly Disagree: 0% of respondents selected these options, indicating no extreme views.

8. Has ICT helped you to become a more independent learner?

Figure 8



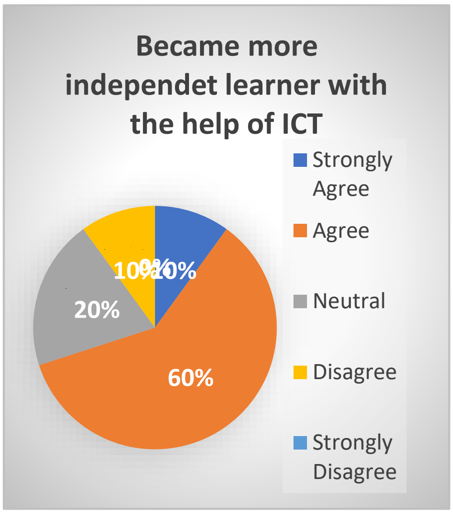


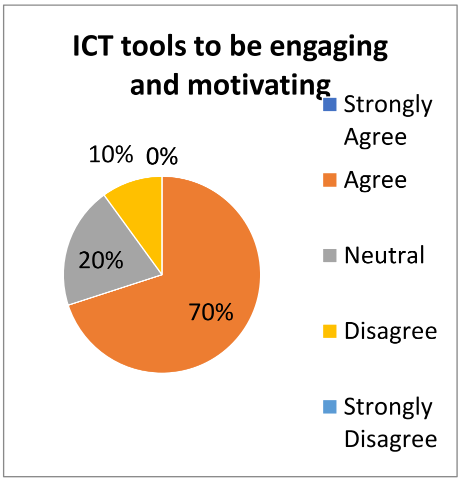
Table 8

Agree: 100% of respondents agree that ICT helps them become more independent learners, making this the unanimous response.	Agree: 60% of respondents agree that ICT helps them become more independent learners, making this the unanimous response.
Neutral: 0% of respondents chose these options, highlighting no disagreement or stronger levels of agreement.	Neutral: 20% of respondents chose these options, highlighting no disagreement or stronger levels of agreement.
Strongly Agree: 0% of respondents chose these options, highlighting no disagreement or stronger levels of agreement.	Strongly Agree: 10% of respondents chose these options, highlighting stronger levels of agreement.

Disagree: 0% of respondents chose these options, highlighting no disagreement or stronger levels of agreement.	Disagree: 10% of respondents chose these options, highlighting disagreement.
Strongly Disagree: 0% of respondents chose these options, highlighting no disagreement or stronger levels of agreement.	Strongly Disagree: 0% of respondents chose these options, highlighting no disagreement.

9. Do you find ICT tools to be engaging and motivating?

Figure 9



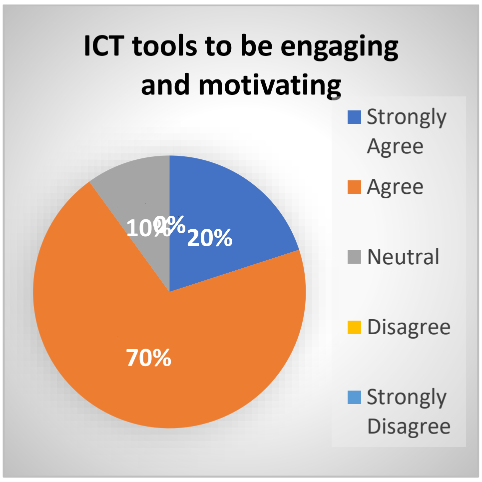


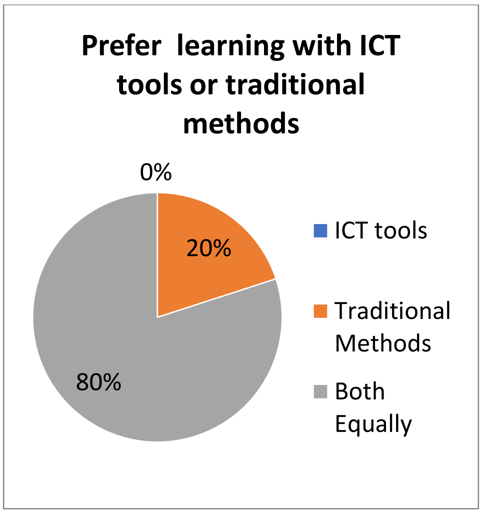
Table 9

Agree: 70% of respondents agree that ICT tools are engaging and motivating, making this the dominant opinion.	Agree: 70% of respondents agree that ICT tools are engaging and motivating, making this the dominant opinion.
Neutral: 20% of respondents are neutral, neither agreeing nor disagreeing.	Neutral: 10% of respondents are neutral, neither agreeing nor disagreeing.

Disagree: 10% of respondents disagree with this statement, reflecting some dissatisfaction.	Disagree: 10% of respondents disagree with this statement, reflecting some dissatisfaction.
Strongly Agree: 0% of respondents expressed these views, indicating no extreme opinions.	Strongly Agree: 20% of respondents expressed these views, indicating strongly agree opinions.
Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.	Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.

10. Do you prefer learning with ICT tools or traditional methods?

Figure 10



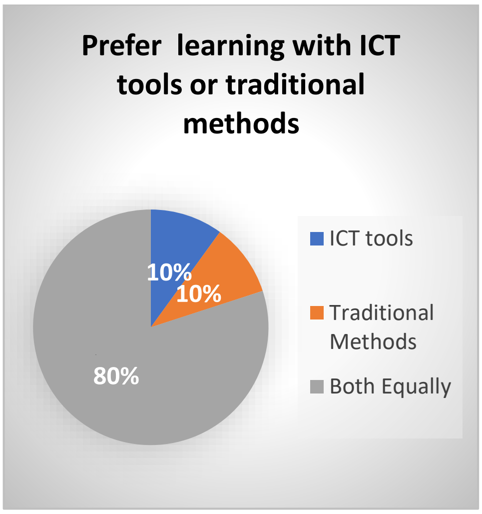


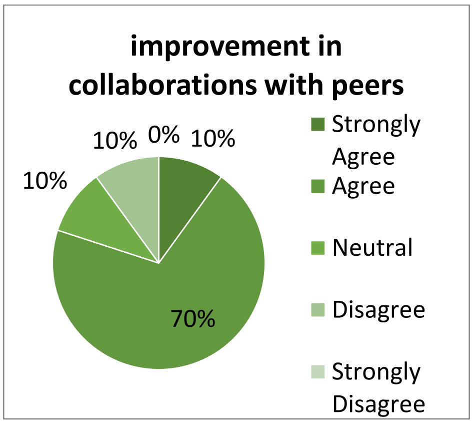
Table 10

ICT Tools: 0% of respondents indicating no extreme opinions on learning from using ICT and traditional methods	ICT Tools: 10% of respondents prefer only ICT tools for learning

Traditional Methods: 20% of respondents prefer traditional way for learning	Traditional Methods: 10% of respondents prefer traditional way for learning
Both Equally: 80% of respondents prefer both ICT tools and traditional methods for learning	Both Equally: 80% of respondents prefer both ICT tools and traditional methods for learning

11. Has ICT improved your ability to collaborate with peers?

Figure 11



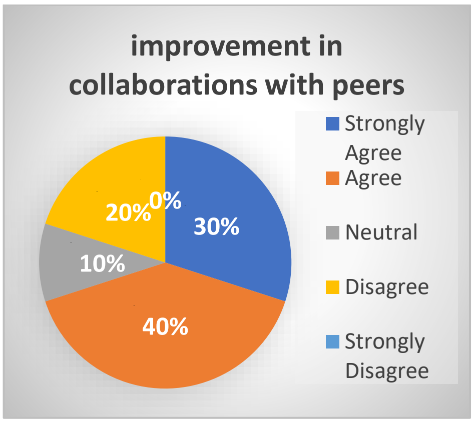


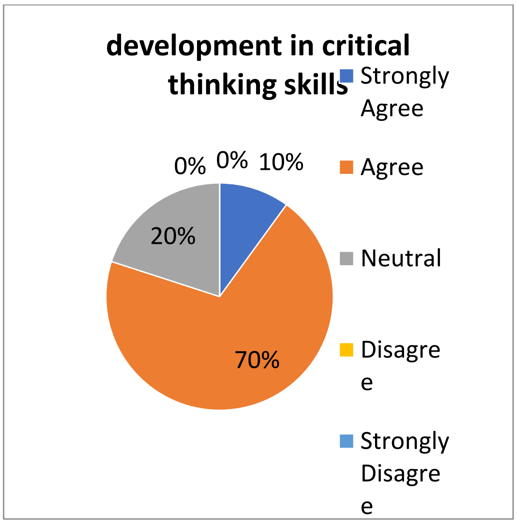
Table 11

Agree: 70% of respondents agree that ICT tools are engaging and motivating, making this the dominant opinion.	Agree: 40% of respondents agree that ICT tools are engaging and motivating, making this the dominant opinion.
Neutral: 10% of respondents are neutral, neither agreeing nor disagreeing.	Neutral: 10% of respondents are neutral, neither agreeing nor disagreeing.

Disagree: 10% of respondents disagree with this statement, reflecting some dissatisfaction.	Disagree: 20% of respondents disagree with this statement, reflecting some dissatisfaction.
Strongly Agree: 10% of respondents expressed these views, indicating extreme opinions.	Strongly Agree: 30% of respondents expressed these views, indicating strongly agreement.
Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.	Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.

12. Do you think ICT has helped you develop critical thinking skills?

Figure 12



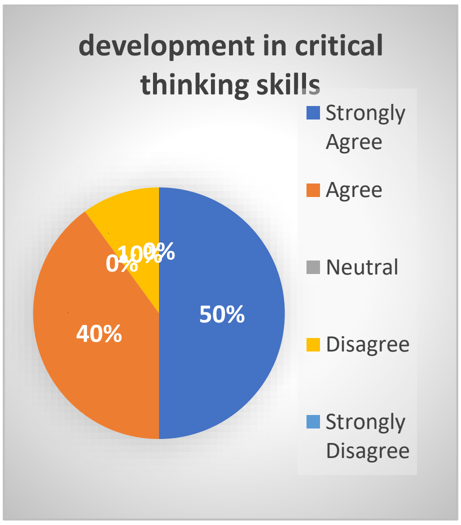


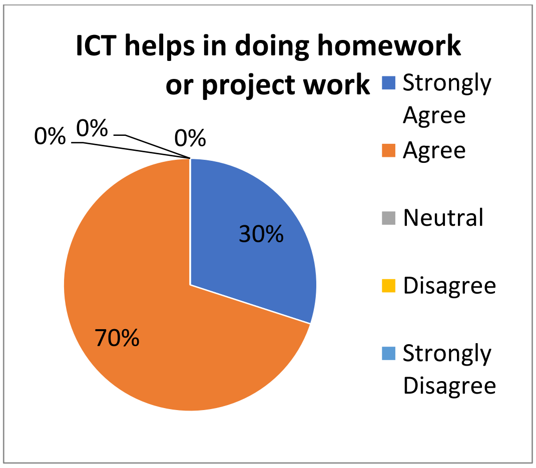
Table 12

Agree: 70% of respondents agree that ICT tools are helping in development in	Agree: 40% of respondents agree that ICT tools are helping in development in

critical thinking skills making this the dominant opinion.	critical thinking skills making this the dominant opinion.
Neutral: 20% of respondents are neutral, neither agreeing nor disagreeing.	Neutral: 0% of respondents are neutral, neither agreeing nor disagreeing.
Disagree: 0% of respondents disagree with this statement, reflecting dissatisfaction.	Disagree: 10% of respondents disagree with this statement, reflecting some dissatisfaction.
Strongly Agree: 10% of respondents expressed these views, indicating strong opinions.	Strongly Agree: 50% of respondents expressed these views, indicating strongly extreme opinions.
Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.	, Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.

13. Has ICT made easier to do homework or projects using ICT tools?

Figure 13



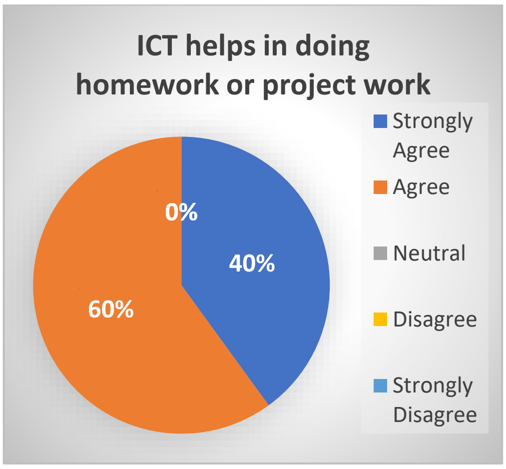
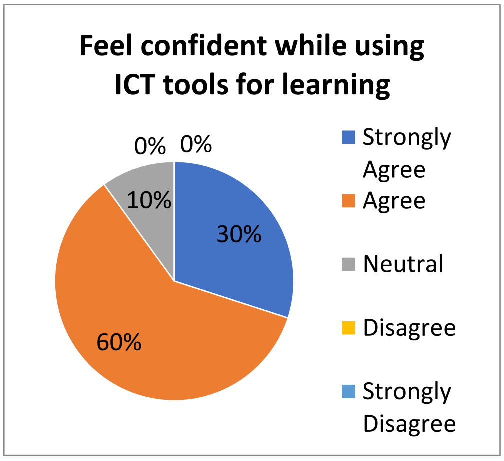


Table 13

Agree: 70% of respondents agree that ICT tools helps in doing homework or project work.	Agree: 60% of respondents agree that ICT tools helps in doing homework or project work.
Neutral: 0% of respondents are neutral, neither agreeing nor disagreeing.	Neutral: 0% of respondents are neutral, neither agreeing nor disagreeing.
Disagree: 0% of respondents disagree with this statement, reflecting dissatisfaction.	Disagree: 0% of respondents disagree with this statement, reflecting dissatisfaction.
Strongly Agree: 30% of respondents expressed these views, indicating strongly agree opinions.	Strongly Agree: 40% of respondents expressed these views, indicating strongly agree opinions.
Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.	Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.

14. Do you feel confident in using ICT tools for learning?

Figure 14



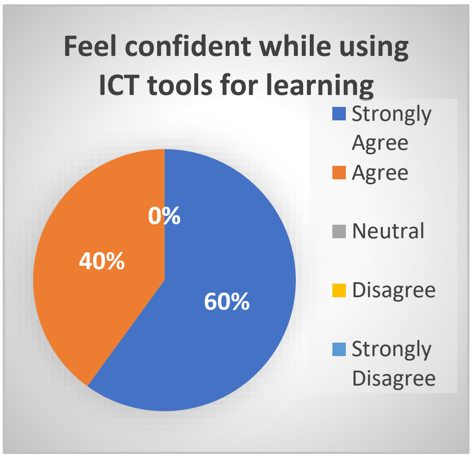


Table 14

Agree: 60% of respondents agree that using ICT tools feel confident for learning.	Agree: 40% of respondents agree that using ICT tools feel confident for learning.
Neutral: 10% of respondents are neutral, neither agreeing nor disagreeing.	Neutral: 0% of respondents are neutral, neither agreeing nor disagreeing.
Disagree: 0% of respondents disagree with this statement, reflecting some dissatisfaction.	Disagree: 0% of respondents disagree with this statement, reflecting some dissatisfaction.
Strongly Agree: 30% of respondents expressed these views, indicating extreme opinions.	Strongly Agree: 60% of respondents expressed these views, indicating extreme opinions.
Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.	Strongly Disagree: 0% of respondents expressed these views, indicating no extreme opinions.

CHAPTER FIVE

5.2 FINDINGS

1. ICT Usage Patterns

Frequency of Computer Usage:

- **Primary Students:** 50% use computers daily, 30% weekly, and 20% monthly. None reported rare or no usage.
- **Secondary Students:** Only 20% use computers daily, while 40% use them weekly, 30% monthly, and 10% rarely.

Finding: Primary students report more consistent daily use of computers, while secondary students have a more varied usage pattern.

Frequency of Smartphone/Tablet Usage:

- **Primary Students:** 50% use these devices daily, 40% a few times weekly, and 10% monthly.
- **Secondary Students:** 70% use them daily, with no monthly users, reflecting heavier reliance on these devices for learning.

Finding: Smartphones and tablets are more commonly used by secondary students, reflecting greater integration into their learning activities.

2. Engagement with ICT Tools

Preferred ICT Tools for Learning:

- **Primary Students:** 50% rely on e-books, 20% use educational apps and online platforms, while only 10% prefer educational websites.
- **Secondary Students:** 50% favor educational websites, followed by 30% for apps, while e-books and platforms are less preferred.

Finding: Primary students prefer e-books for learning, while secondary students focus more on interactive educational websites.

3. Impact on Learning Outcomes

Understanding of Subjects:

- Both groups agree ICT improves understanding, but secondary students show stronger agreement (40% "strongly agree" compared to 0% for primary students).

Finding: ICT tools are perceived as effective for improving subject understanding, with a stronger impact noted among secondary students.

Independence in Learning:

- **Primary Students:** All agree ICT helps them become independent learners.
- **Secondary Students:** 60% agree, with 10% strongly agreeing and 10% disagreeing.

Finding: Primary students unanimously believe ICT fosters independence, while secondary students have a more nuanced view.

Collaboration Skills:

- **Primary Students:** 70% agree ICT enhances collaboration.
- **Secondary Students:** 40% agree, with 30% strongly agreeing and 20% disagreeing.

Finding: ICT has a notable impact on collaboration for both groups, with slightly stronger opinions among secondary students.

4. Development of Critical Thinking Skills

- **Primary Students:** 70% agree, with 10% strongly agreeing, while 20% are neutral.
- **Secondary Students:** 40% agree, with 50% strongly agreeing, and 10% disagreeing.

Finding: Both groups recognize ICT's role in fostering critical thinking, but secondary students report stronger agreement.

CHAPTER SIX

6.1. CONCLUSION

This study has explored the transformative impact of Information and Communication Technology (ICT) on education, focusing on how ICT influences the academic performance and learning experiences of primary and secondary-level students in schools located in Vyas-2, Damauli, Tanahun. The findings reveal that ICT tools play a crucial role in shifting traditional teacher-centered classrooms to student-centered learning environments, enhancing engagement, motivation, and collaboration among students. However, the study also highlights distinct differences in how primary and secondary students interact with and benefit from ICT tools.

For primary students, ICT tools such as e-books, educational apps, and multimedia presentations are particularly effective in fostering engagement and motivation. These tools provide intuitive and guided learning experiences that align with the developmental needs of younger learners. Primary students rely heavily on teacher support to navigate ICT, and while they show improvements in collaboration and engagement, their ability to independently explore and critically analyze content remains limited. The structured use of ICT in primary education supports the development of foundational skills and prepares students for more advanced applications of technology in their academic journey.

Secondary students, in contrast, demonstrate a higher level of independence and critical thinking when utilizing ICT tools. They frequently engage with advanced technologies such as educational websites, online learning platforms, and collaborative tools, which enhance their ability to explore complex subjects, retain knowledge, and solve problems. Secondary students also exhibit greater confidence in using ICT for learning, attributing this to their advanced cognitive development, familiarity with technology, and the broader scope of subjects they study. ICT tools enable secondary students to transition from passive recipients of information to active participants in their education, fostering critical thinking, collaboration, and self-directed learning.

The comparative analysis underscores the need for differentiated ICT integration strategies that cater to the unique needs of learners at different stages. For primary students, the focus should be on creating intuitive, interactive, and teacher-guided ICT experiences that build engagement and foundational skills. For secondary students, educational institutions should emphasize providing access to advanced ICT tools and fostering an environment that encourages independent exploration and critical thinking.

Moreover, the study highlights the importance of context-specific research, as the findings from schools in Vyas-2, Damauli, may not be universally applicable. The research's localized nature emphasizes the need for tailored ICT strategies that consider geographic, institutional, and cultural factors.

In conclusion, ICT has the potential to bridge educational gaps and provide equitable learning opportunities across grade levels. However, its effectiveness depends on thoughtful implementation that aligns with the developmental stages, cognitive abilities, and learning needs of students. By leveraging the strengths of ICT and addressing its challenges, educators and policymakers can ensure that technology serves as a transformative force in education, empowering students to achieve their full potential in an increasingly digital world.

6.2. RECOMMENDATIONS

Tailored ICT Integration: Tailored ICT integration is essential for meeting the diverse needs of students across different age groups. For primary students, interactive tools like educational games and multimedia resources can enhance engagement and foundational learning, while secondary students benefit from advanced tools that foster critical thinking and problem-solving skills.

Teacher Training: Teacher training should focus on equipping educators with the skills to effectively integrate ICT into their teaching practices, ensuring they are confident in using collaborative and interactive methods.

Infrastructure Development: Infrastructure development is equally vital, with schools needing reliable access to ICT tools, internet connectivity, and devices. Resource sharing among institutions can help bridge gaps in availability and create a more balanced educational landscape.

Curriculum Updates: Updating the curriculum to integrate ICT across all subjects and including ICT literacy as a core component ensures that students are well-prepared for a technology-driven world.

Parental Involvement: Parental involvement also plays a key role, with awareness programs designed to encourage the use of ICT at home and equip parents to support their children's learning.

Monitoring and Evaluation: Continuous monitoring and evaluation of ICT’s impact on learning outcomes are necessary, enabling strategies to evolve based on stakeholder feedback.

Equitable Access: Ensuring equitable access remains a top priority, addressing barriers such as device shortages and training gaps through targeted policies to reduce the digital divide, particularly for disadvantaged students.

Collaborative Learning: Collaborative learning can be promoted by leveraging digital platforms for group projects and peer-to-peer interactions, fostering teamwork and enhancing the overall learning experience.

CHAPTER SEVEN

7.1. REFERENCES

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CHAPTER EIGHT

8.1. QUESTIONNARE

Questionnaires on the Comparative Impact of ICT on Student Learning Outcomes

1. What is your age?

   ____

2. What is your grade Level?

   ____

3. What is your school name?

   ____

4. How often do you use a computer at school?

   i. Daily
   ii. Weekly
   iii. Monthly
   iv. Rarely
   v. Never

5. How often do you use a smartphone or tablet for educational purposes?

   i. Daily
   ii. Weekly
   iii. Monthly
   iv. Rarely
   v. Never

6. What ICT tools do you use most frequently for learning?

   i. Educational websites
   ii. Online learning platforms

iii. E-books

iv. Educational apps

v. Video conferencing tools

vi. Other (please specify)

____

7. Do you think ICT tools have improved your understanding of subjects?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

8. Has ICT helped you to become a more independent learner?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

9. Do you find ICT tools to be engaging and motivating?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

10. Do you prefer learning with ICT tools or traditional methods?

i. ICT tools

ii. Traditional methods

iii. Both equally

11. Has ICT improved your ability to collaborate with peers?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

12. Do you think ICT has helped you develop critical thinking skills?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

13. ICT made easier to do homework or projects using ICT tools?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

14. Do you feel confident in using ICT tools for learning?

i. Strongly Agree

ii. Agree

iii. Neutral

iv. Disagree

v. Strongly Disagree

CHAPTER NINE

9.1. PHOTO GALLERY